COURSE CODE
SBT-DF202

COURSE TITLE
Computer and Digital Forensics

LAB TITLE
USB Image Acquisition and Hash Verification

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STUDENT ID
2025/FWSD/11325

DATE
30/08/2026

PLATFORM USED
Kali Linux

INSTRUCTOR
Mr. Aminu Idris, AMCPN
Commandant

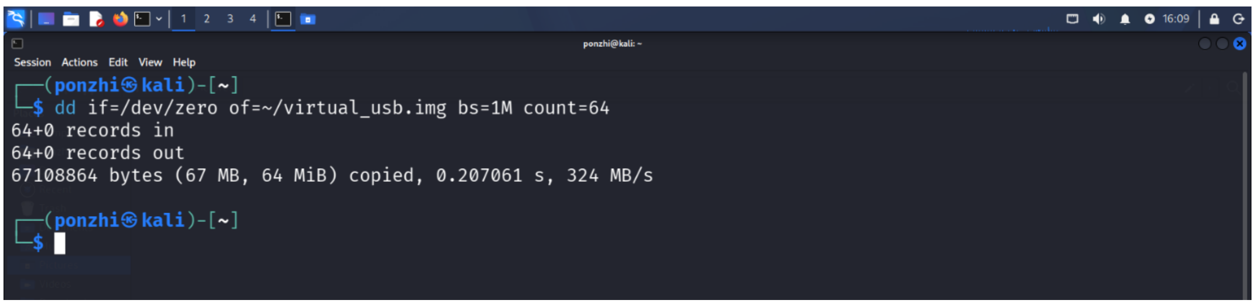
Lab review

By completing this lab, I will demonstrate the ability to explain the purpose of forensic disk imaging, prepare controlled USB evidence, acquire a forensic USB image dd/dcfldd on Linux, calculate MD5 and SHA-256 hash values, compare source and image hashes for integrity verification, validate the contents of the acquired image, identify risks associated with low-level imaging tools, and document the complete acquisition process with professional screenshots and evidence notes.

Linux walkthrough using a loopback virtual disk instead of a physical USB

Create a container file to act as the raw disk

dd if=/dev/zero of=~/virtual_usb.img bs=1M count=64



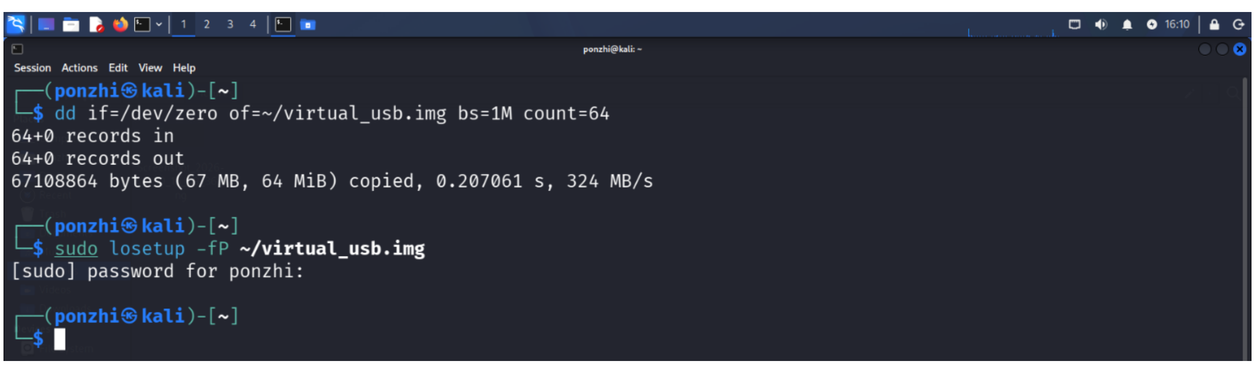
- if=/dev/zero: This input provides an endless stream of zero bytes to fill the space.

- of=~/virtual_usb.img: This is the output file, which will be your virtual disk.

- bs=1M count=64: This command writes in 1MB blocks, a total of 64 times, which makes a 64MB file.

Attach the file as a loop device

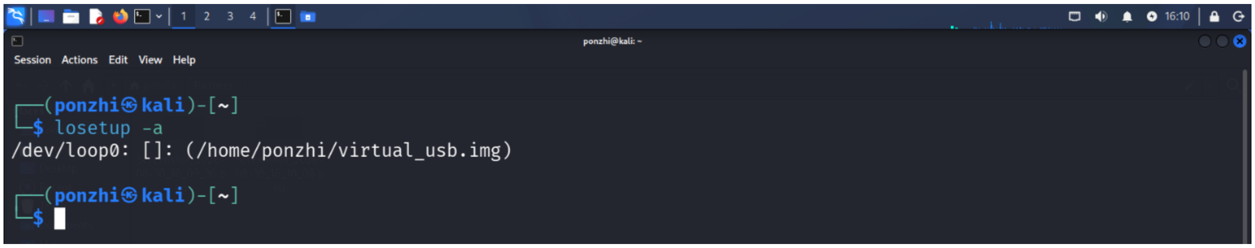
sudo losetup -fP ~/virtual_usb.img



-f: automatically finds the first free loop device.
-P: tells the kernel to scan for partitions on it once one exists.

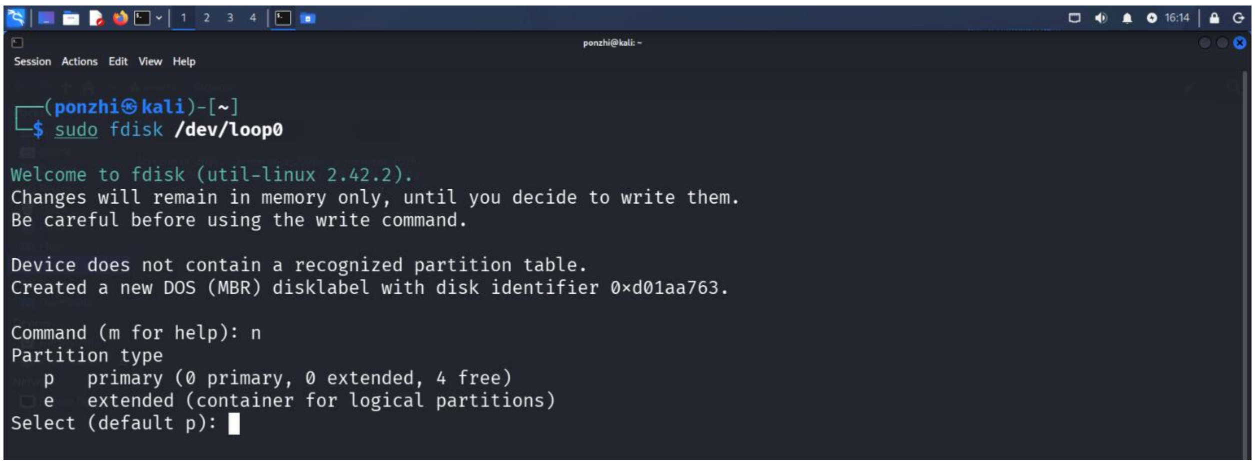
Confirm which loop device you got

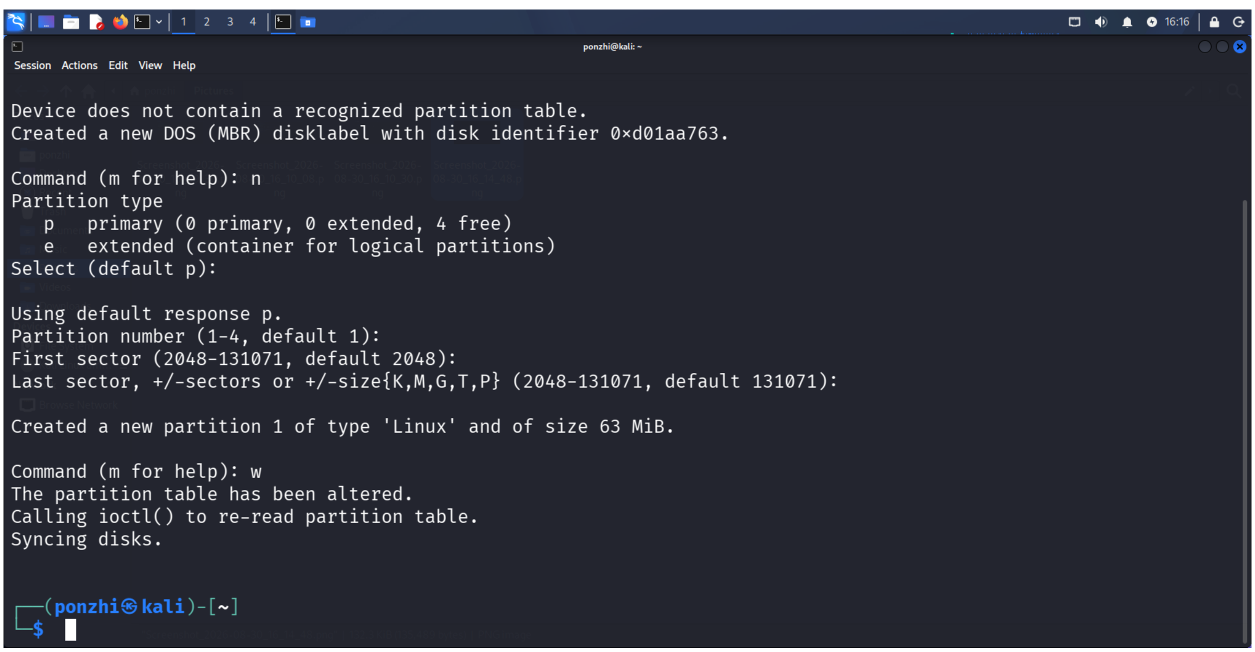
losetup -a



Partition it

sudo fdisk /dev/loop0

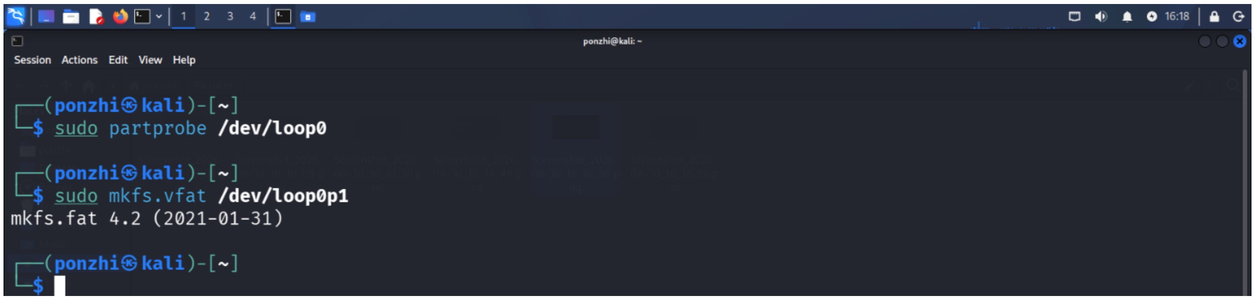




Tell the kernel to re-read the new partition table and format the partition with a filesystem

sudo partprobe /dev/loop0

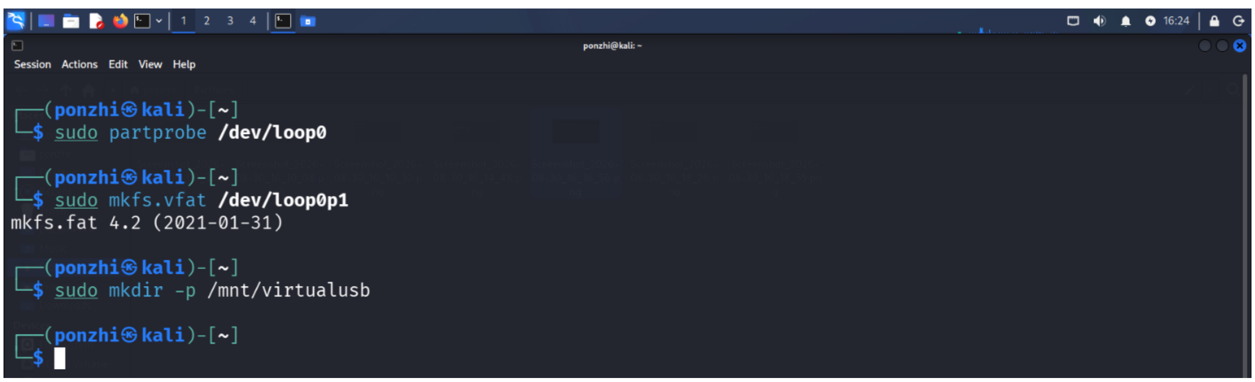
sudo mkfs.vfat /dev/loop0p1



This makes /dev/loop0p1 appear without needing a reboot

Format the partition with a filesystem

sudo mkfs.vfat /dev/loop0p1



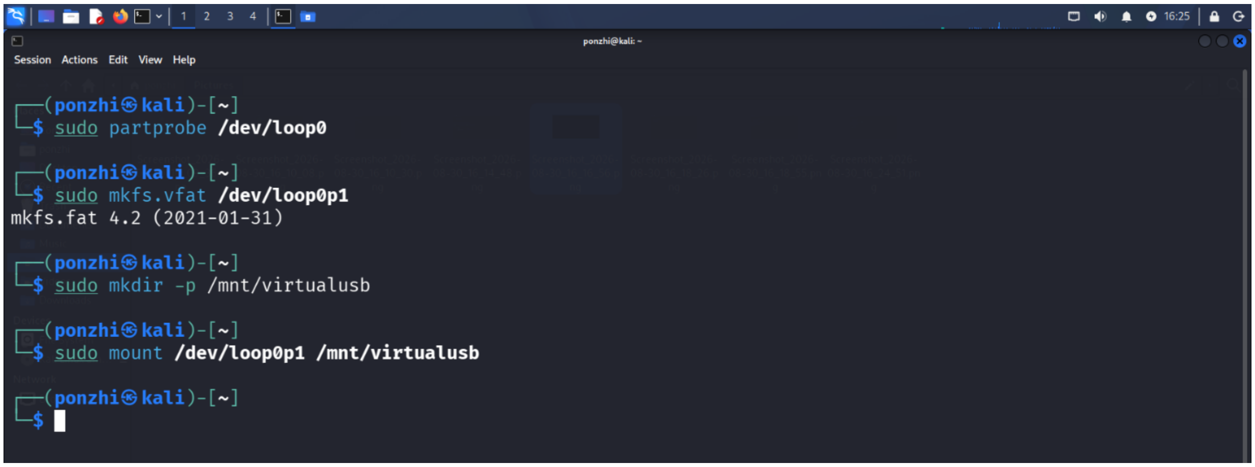
FAT32 mirrors what a typical real USB uses, so my forensic steps closely match the real lab

Part A – Prepare the USB Evidence

Mount the Virtual USB

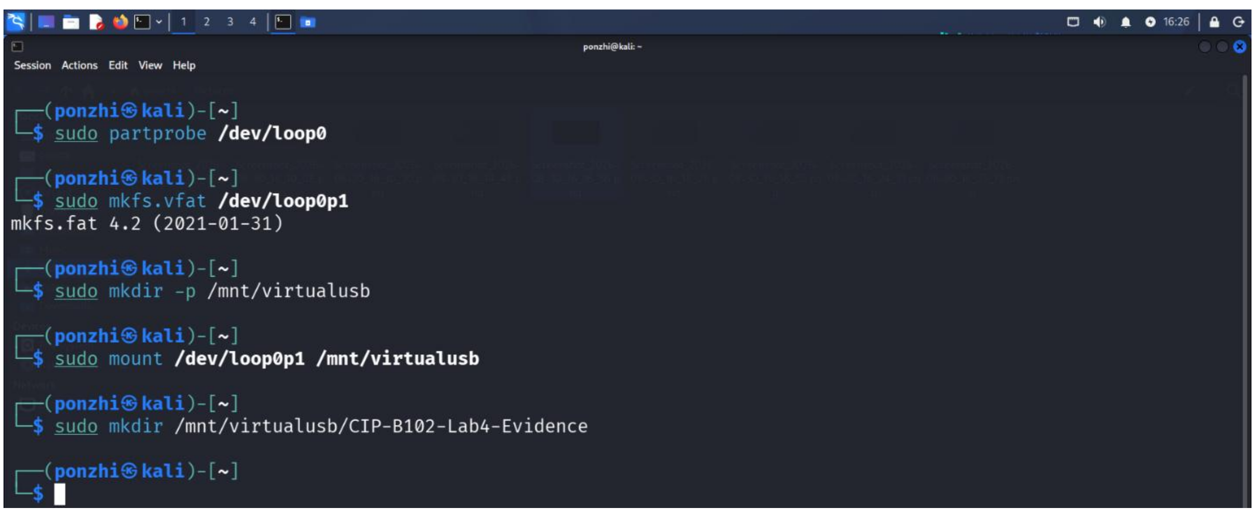
sudo mkdir -p /mnt/virtualusb

sudo mount /dev/loop0p1 /mnt/virtualusb

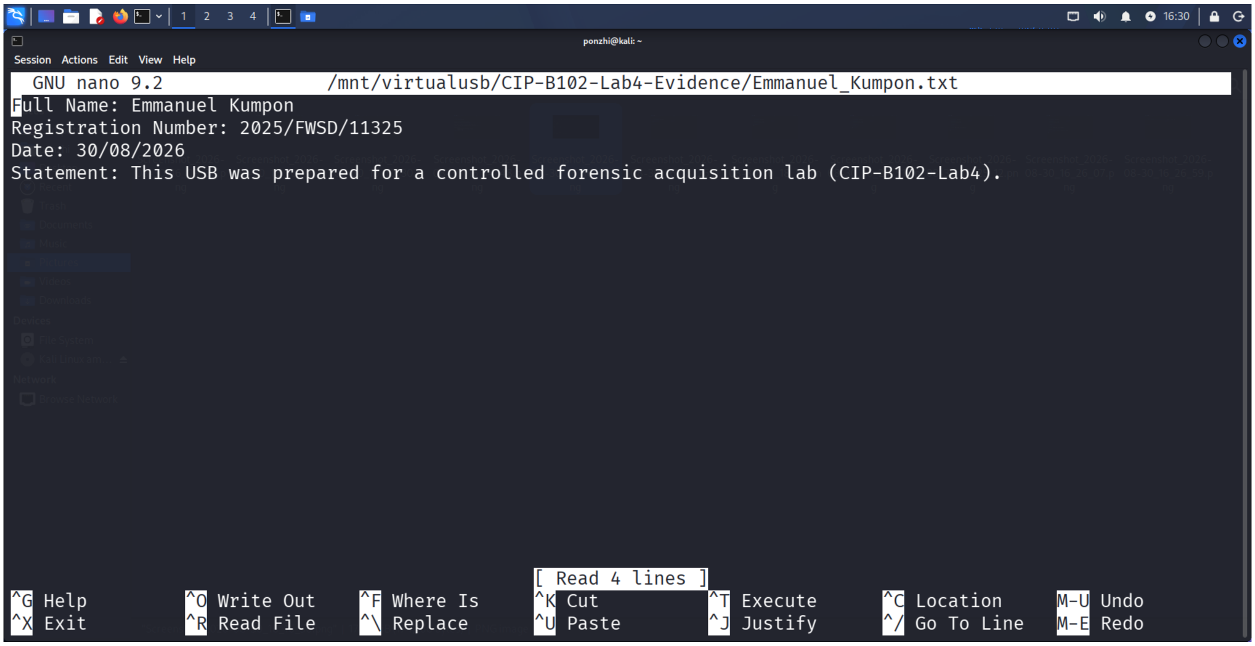


Create the evidence folder and file

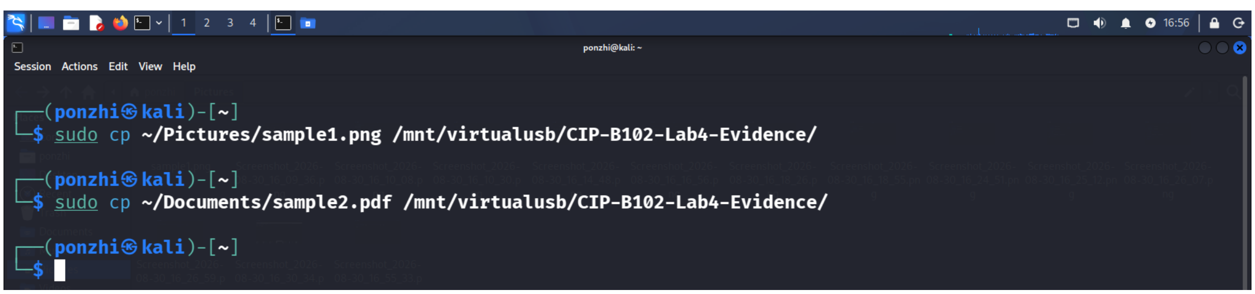
sudo mkdir /mnt/virtualusb/CIP-B102-Lab4-Evidence



sudo nano /mnt/virtualusb/CIP-B102-Lab4-Evidence/Emmanuel_Kumpon.txt

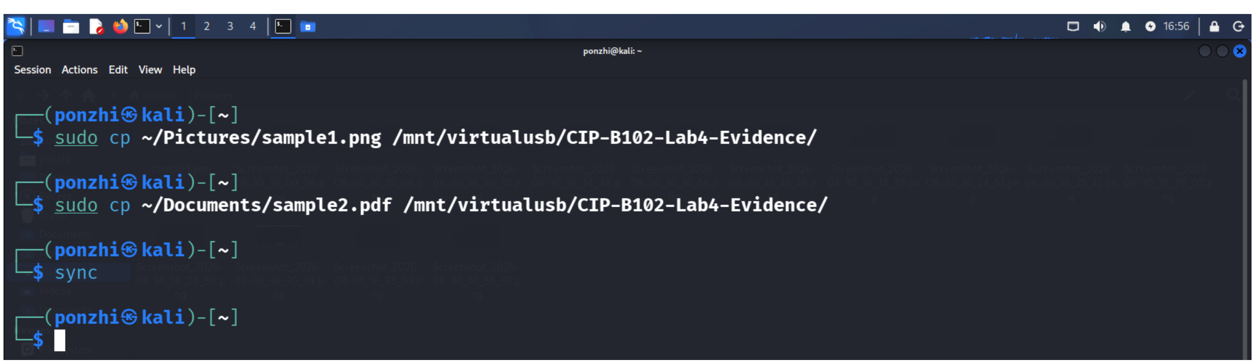


Add one image and one PDF



Flush writes to disk

sync



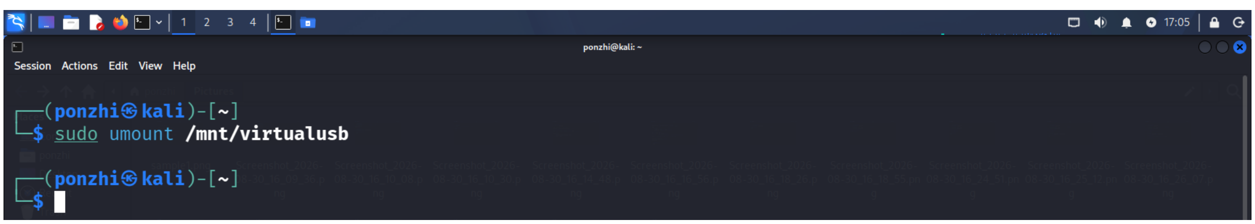
Part B - Complete Evidence Documentation

Case / Lab ID	CIP-B102-Lab4
Student name	Emmanuel Kumpon
USB brand/description	Simulated USB drive (loopback virtual disk, /dev/loop0)
USB capacity	64MB
Acquisition method	dd (Linux), loop device substituted for physical USB
Tool and version	GNU coreutils dd; fdisk
Image filename	CIPB102_Lab4_Emmanuel_Kumpon_USB.dd
Image storage location	~/CIP-B102-Lab4
Hash algorithms used	MD5, SHA-256

Part C - Acquire the USB Image

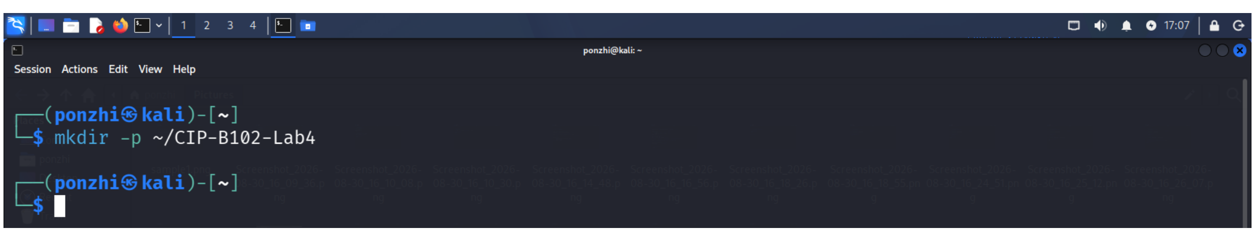
Unmount the partition first

sudo umount /mnt/virtualusb



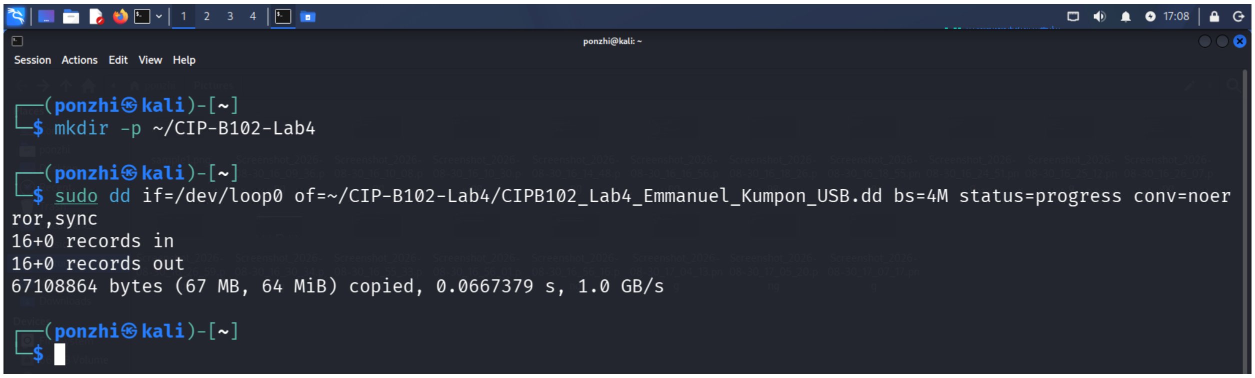
Create the destination folder on your real disk

mkdir -p ~/CIP-B102-Lab4



Run the acquisition image on the whole loop device, not just the partition

sudo dd if=/dev/loop0 of=~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd bs=4M status=progress conv=noerror,sync



if=/dev/loop0: The entire virtual device, the loopback equivalent of /dev/sdb.

bs=4M: 4MB block size for speed.

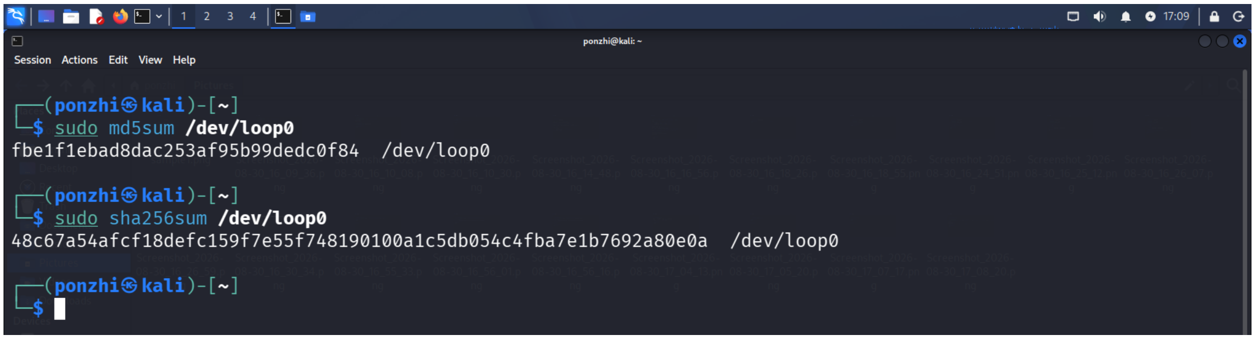
status=progress: live progress display.
conv=noerror, sync: keep going past unreadable blocks, padding with zeros to preserve alignment.

Part D - Hash Verification

Hash the source

sudo md5sum /dev/loop0

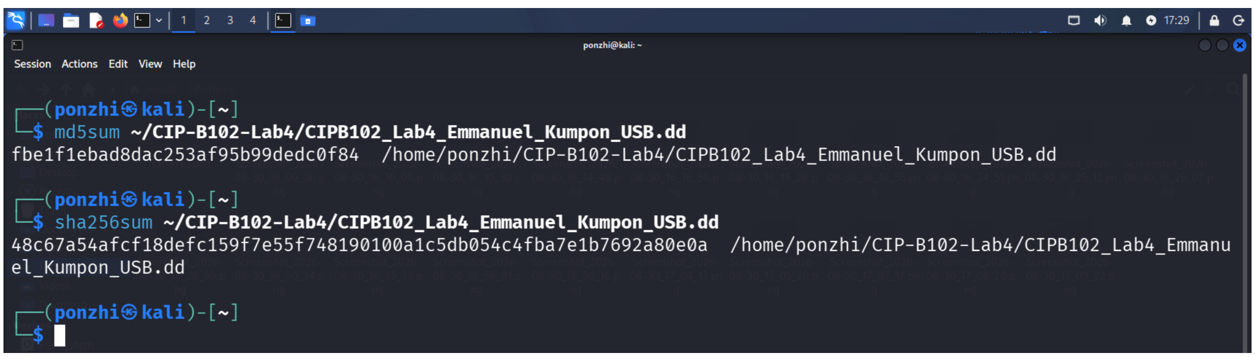
sudo sha256sum /dev/loop0



Hash the acquired image file

md5sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd

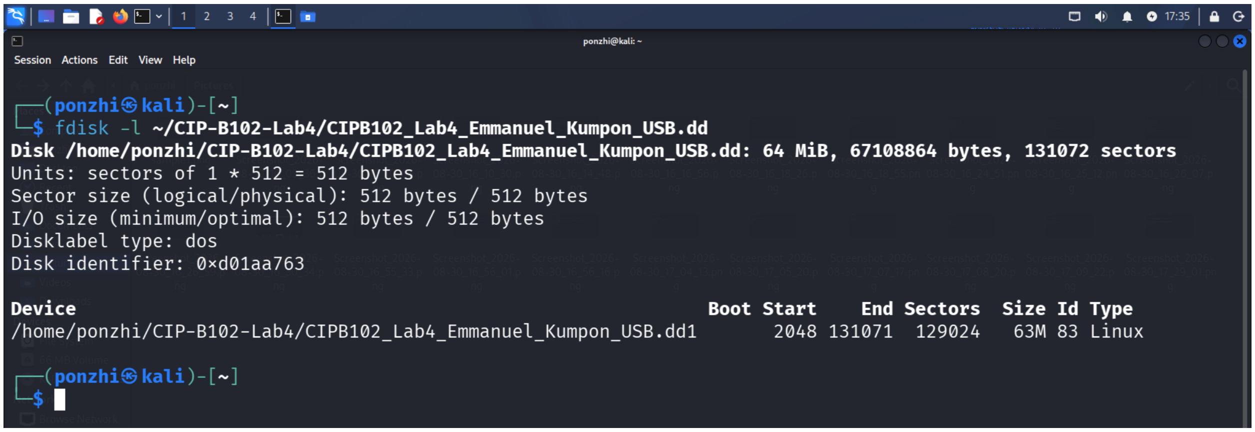
sha256sum ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd



Part E - Validate the Acquired Image

Find the partition's byte offset inside the image

fdisk -l ~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd



Calculate the offset

offset = start_sector × sector_size

start_sector = 2048

sector_size = 512

Offset = 2048 x 512 = 1,048,576

Mount the image read-only through loopback

mkdir ~/image_mount

sudo mount -o ro,loop,offset=1048576 ~/CIP-B102-Lab4/CIPB102_Lab4_Emmanuel_Kumpon_USB.dd ~/image_mount

Confirm the evidence file is present and intact

ls -R ~/image_mount

cat ~/image_mount/CIP-B102-Lab4-Evidence/Emmanuel_Kumpon.txt

Conclusion

We completed the acquisition process using a loopback virtual disk instead of a physical USB device. The process included preparing evidence, documenting steps before the acquisition, creating a raw image with dd, checking the cryptographic hash, and verifying the image content in read-only mode. This process follows the standard procedures for acquiring data from a physical USB and shows that we have the necessary skills for this lab.